		Potential energy change = 50 [–70 – (–30)] = –2000 MJ
13	**D**	A is true: From $pV = nRT$, at constant pressure, when volume is doubled, temperature is doubled. B is true: work done $W = p\Delta V = 4P_o(2V_o - V_o)$ C is true: $W = 0 \rightarrow \Delta U = Q$. From $pV = nRT$, since there is a decrease in pressure, there must be a decrease in temperature and hence a decrease in internal energy, i.e. ΔU is negative. This means Q is negative, i.e. heat is removed.
14	**C**	From $pV = nRT$, when p and V are tripled, the temperature would be $9T$